

ETHGas and the “Gasless Ethereum” Future – First Defense Research Report

What is ETHGas? Explaining Blockspace Futures in Simple Terms

ETHGas is a new Ethereum infrastructure project that aims to make **Ethereum “blockspace” tradeable** – essentially letting people buy and sell the right to use space in future Ethereum blocks ¹. In other words, ETHGas is creating a **futures market for Ethereum gas fees**. By doing so, it claims it can tame gas price volatility and even enable a “**gasless Ethereum**” experience for users ².

- **Ethereum Blockspace Futures:** In traditional markets, a futures contract lets you lock in the price of a commodity (like oil or grain) for future delivery. Similarly, an Ethereum gas futures market would let users and developers lock in **today’s price for future blockchain usage** ³. Vitalik Buterin (Ethereum’s co-founder) recently called for a “trustless on-chain gas futures market” to help everyone hedge and plan for transaction costs ⁴ ³. The idea is that instead of worrying “*gas fees are low now, but what about in two years?*”, large users could prepay or secure **blockspace** in advance ⁵. This provides a clear signal of expected future fees and allows hedging against spikes ⁶. **Blockspace futures** turn the cost of Ethereum transactions into a predictable asset, much like airlines buy fuel futures to stabilize costs ⁷.
- **What ETHGas Is Building:** ETHGas is creating the **infrastructure and marketplace** to trade these blockspace futures. It’s essentially a specialized exchange where Ethereum validators (the block producers) can sell commitments for future block capacity, and buyers (like apps, traders, or even other block builders) can purchase that capacity ahead of time ⁸. The platform raised a \$12 million seed round led by Polychain Capital and launched with about **\$800 million in blockspace purchase commitments** from leading Ethereum builders, validators, and other participants ⁹ ¹⁰. In simpler terms, many key players have agreed to use or support this system, signaling significant interest. ETHGas describes itself as a settlement layer that **makes Ethereum’s computation and storage “pre-bookable”** – users can acquire, trade, or hedge *any amount of future transaction capacity* on the blockchain ¹.
- **“Gasless Ethereum” in Practice:** Despite the buzzword, “*gasless*” *doesn’t literally mean no gas fees exist*. Instead, it means **end-users wouldn’t directly pay gas** when using Ethereum. ETHGas envisions applications or protocols covering the gas costs on users’ behalf, using the blockspace market to manage those costs efficiently ². For example, a decentralized app (dApp) could **pre-purchase cheap blockspace futures** and use those to run transactions for its users, who then experience the dApp without paying any gas at the point of use. Gas fees become “**invisible**” **to the consumer** ¹¹ ¹². This is achieved via programs like ETHGas’s **Open Gas Initiative (OGI)**, which allows protocols to sponsor user gas fees easily. Already, OGI has enabled over 127,000 users to get **gas fee rebates or sponsorships** – collectively identifying over 270,000 ETH worth of historical gas spend that dApps could reimburse ¹³ ¹⁴. In plain terms, ETHGas turns gas fees into a **marketing and user-retention tool**: projects can pay your transaction fee to attract you as a user ¹³. Thus, “gasless Ethereum” means your wallet might say \$0.00 gas, while the fee is paid behind the scenes by the app or a sponsor (who likely budgeted for it or hedged it in the futures market).

Why does this matter? Ethereum gas fees have a history of being **volatile and unpredictable**, which hurts user experience. During peak demand, average transaction fees have spiked to painful levels (over \$50 at times) and then dropped to low dollars at others ¹⁵. Even as Ethereum has scaled and fees hit multi-year lows in 2025, users and businesses worry that a new surge in usage (or a popular app) could send gas costs soaring again ¹⁵. The chart below illustrates Ethereum's gas price volatility over time, which blockspace futures aim to mitigate:

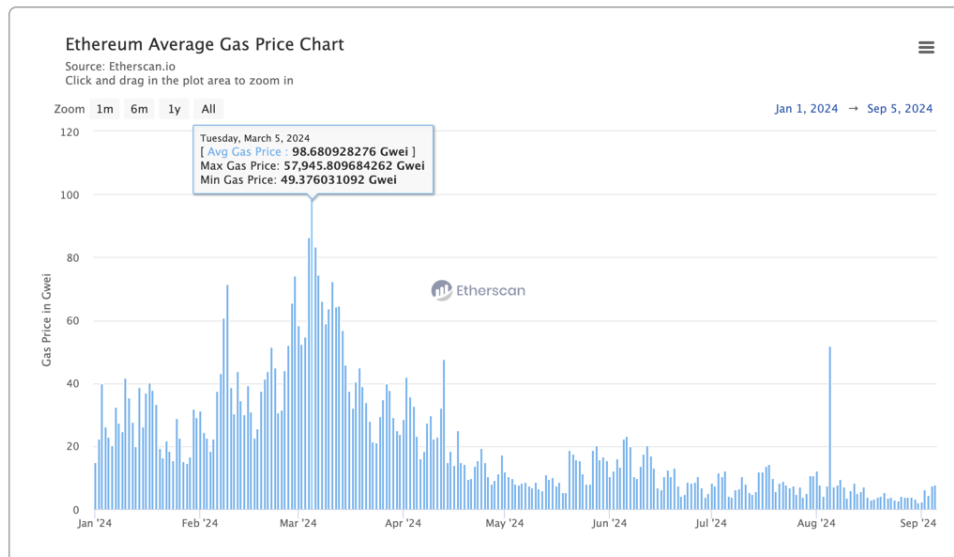


Chart: Ethereum's average gas price (in gwei) from early 2024 through late 2024, showing significant volatility and a downward trend after mid-2024 upgrades. Such fee unpredictability is what ETHGas hopes to address by letting participants lock in gas costs in advance ¹⁶ ¹⁵.

By making gas an asset you can trade or reserve, ETHGas claims Ethereum can evolve into a **“real-time network” with instant confirmations and stable costs** ² ¹⁷. In summary, ETHGas is building a kind of *gas fee booking system*: you reserve block space now so you're not at the mercy of whatever the market price is later. And if done right, you might never have to fiddle with a gas slider in your wallet again – the dApp or wallet would handle it, giving you a **gas-free feel** (even though the fees are just being paid elsewhere).

How ETHGas Works: Blockspace Markets and Core Mechanisms

Blockspace as a Market: Under the hood, ETHGas turns Ethereum's transaction fee auction (which currently happens on the fly each block) into a more structured market. Instead of blindly bidding every time you send a transaction, participants can **buy rights to blockspace in advance**. ETHGas introduces the concept of **“blockspace commitments” or “pre-confirmations.”** These are basically promises that a certain transaction *will be included* at a certain time or within a certain future block, because the right to that block space has been pre-purchased ¹⁸. Think of it like reserving a spot in an upcoming block – you get a **confirmation ahead of time** (“pre-confirmation”) that your transaction is guaranteed to execute at block X or within time window Y.

- **How It Might Be Implemented:** ETHGas works closely with the Ethereum validator and block builder ecosystem to enable this. When a validator knows it will be producing a block in the near future (Ethereum rotates proposers each slot ~ every 12 seconds), ETHGas allows the validator to **sell that upcoming block's space** via an auction on the ETHGas exchange ¹⁹. Buyers (which could be high-volume dApps, traders, or even specialized “builders”) bid for either whole blocks

or just slices of blockspace (“inclusion slots”). If a purchase is made, the buyer gets the **sequencing rights** to that block – meaning they can decide what transactions go in and in what order ¹⁹ ²⁰. The validator, in turn, gets paid a predictable fee for the block in advance (likely higher or more stable than just hoping for random user tips).

To ensure trust, ETHGas requires validators to put up **collateral** and uses a slashing mechanism. Validators run a sidecar software called **Commit Boost** that signals their intent to sell blockspace and signs these commitments ²¹. If a validator sells a future slot and then fails to include the promised transactions (for example, if they go offline or try to cheat), their posted collateral (e.g. 1 ETH per slot) can be **slashed (seized)** and given as compensation to the affected buyers ²² ²³. This creates a financial incentive for block producers to honor all pre-sold commitments. Conversely, ETHGas itself posts a liveness bond to cover failures on the platform side ²⁴. In short, cryptoeconomic guarantees back every reserved transaction – either the block gets delivered as promised, or someone pays a penalty.

- **Pre-Purchasing & Gas Tokenization:** In ETHGas’s system, blockspace becomes a token-like commodity. While earlier attempts at “gas token” contracts (like **GST2 or CHI tokens**) tried to let users **store cheap gas for later** (by exploiting gas refunds for storage) ²⁵, those hacks were rendered obsolete by Ethereum protocol changes (EIP-3529 removed most refunds, essentially killing gas tokens) ²⁶. ETHGas approaches gas tokenization more directly: rather than caching gas, you **purchase a contract that entitles you to X amount of gas in a future block**. This can be thought of as tokenizing blockspace – turning ephemeral capacity into a standardized product. For instance, ETHGas plans to introduce “**Base Fee futures**,” which are like forward contracts on the Ethereum base fee for a specific block slot ²⁷. Traders could go long or short on the base fee level itself as a way to hedge or speculate on gas prices, with cash settlement in ETH ²⁸. These instruments parallel traditional derivatives (imagine betting on gas prices in June 2026, for example).

More concretely, ETHGas’s marketplace offers different products: - **Inclusion Pre-confirmations:** Reserve a spot for a specific transaction (or bundle of transactions) in an upcoming block. This is useful for latency-sensitive actions – e.g. a DeFi trade or NFT mint that you want to ensure goes through at a set time ²⁹. - **Execution Pre-confirmations:** Likely guarantees that a certain *execution payload* or sequence of actions will happen (geared toward algorithmic traders or arbitrage bots that need deterministic execution) ²⁹. - **Whole Block Commitments:** Buy an entire block’s worth of space in advance ²⁹. This would appeal to large arbitrageurs or rollup sequencers who may occasionally want a whole block (for instance, to post a massive batch of L2 transactions or execute a complex multi-step trade without competition). It inherently carries more counterparty risk (since you’re dealing with an entire block and the validator’s honesty), but also more control. - **Blockspace Futures & Options:** As noted, ETHGas intends to allow purely financial trading on gas price indices (base fee futures) in Q1 2026 ²⁷. These would be *cash-settled* (no actual blockspace delivery, just payout based on price) and require margin like any derivative ³⁰.

ETHGas essentially builds a **parallel pipeline alongside Ethereum’s normal fee mechanism** to support these. It integrates with **Proposer-Builder Separation (PBS)** – the system where validators outsource block construction to expert builders. ETHGas’s exchange can be seen as a **pre-PBS auction**: e.g. a validator delegates its slot to ETHGas’s market, traders bid and maybe a “Realtime builder” wins. This realtime block builder then constructs the block (including all pre-sold transactions) and submits it via the normal relay to the validator for final on-chain inclusion ¹⁹ ²⁰. During the block slot, that builder also streams **micro-block updates** to users (more on that shortly).

Comparison to Existing Gas Solutions: ETHGas’s blockspace market approach is novel, but it’s useful to contrast it with existing mechanisms aimed at managing gas costs and user experience:

Solution	How It Works	Pros	Cons / Limits
Ethereum Today (EIP-1559)	<i>Base fee</i> burned per block, users add tips; block size flexes to target 50% usage ³¹ .	More stable than first-price auctions (pre-2021); base fee burn makes ETH deflationary ³² .	Cannot lock or predict fees long-term; fees still spike with demand – no built-in hedging.
Layer-2 Rollups	Transact on L2 networks (Optimism, Arbitrum, etc.) which batch many tx's into one L1 tx. Users pay very low fees in L2 token; L2 posts compressed data to Ethereum.	Much lower user fees (90%+ reduction in many cases after upgrades) ³³ ; higher throughput. Shields users from L1 fee spikes most of the time.	Not “free” – rollup operators still pay L1 gas to post data (cost passed indirectly to users). If L1 fees surge, it eventually affects L2 costs. Also, different trust assumptions (centralized sequencers, etc. in many cases).
Gas Tokens (Now obsolete)	Smart contracts (e.g. GST2, CHI) that let users <i>pre-buy gas</i> by doing wasteful operations when gas is cheap, then reclaim that gas later via refunds when gas is expensive ²⁵ .	Clever way to arbitrage gas price swings; allowed saving on expensive txs by burning tokens bought during cheap periods.	Ethereum ended gas refunds in 2021 (EIP-3529) – making these tokens worthless ²⁶ . Even before that, it was technical and mainly benefited advanced users and bots. Not a long-term UX solution.
ETHGas Blockspace Futures	Pre-book block capacity or price: either get a guaranteed slot for your transactions (“pre-confirmation”) or lock in a gas price via a futures contract. Conducted via off-chain marketplace integrated with validators ⁸ ¹⁸ .	Predictable costs (hedge against volatility); ability to guarantee fast inclusion (no waiting or bidding war); end-users can enjoy “gasless” UX as apps manage gas behind the scenes ² . Also potentially improves validator revenues and staking yields ³⁴ ³⁵ .	Complexity – adds financial markets to blockspace (requires collateral, risk management, technical integration with validators). Still experimental: needs enough participants to be liquid and effective. May introduce new risks (discussed later) such as market manipulation or centralization of block building.

Table: A comparison of ETHGas's approach with EIP-1559 (the current fee mechanism), Layer-2 scaling, and the old gas token method.

Precision and “Real-Time” Ethereum: One notable promise of ETHGas is ultra-fast, predictable execution. By coordinating transactions ahead of time, ETHGas claims it can deliver **sub-second finality** for transactions in practice ³⁶ ³⁷. How? Through what they call **Realtime Blocks**. Normally, Ethereum finalizes blocks every ~12 seconds, which means users wait up to 12s (often longer) to see a transaction confirmed. ETHGas's realtime block builder breaks a block into **hundreds of micro-intervals**, updating the pending state continuously (e.g. every 50–100 milliseconds) and broadcasting these “**pre-confirmations**” to users via participating RPC providers ³⁸ ³⁹. In effect, users and dApps connected to an ETHGas-enabled node can see transaction results almost instantly (within 50–100ms) instead of

waiting one whole block. The final on-chain block still comes at the 12s mark, but by then it's just confirming what was already previewed. This approach is like showing *incremental block updates* in real-time, giving the **illusion of a 50ms block time** without actually altering Ethereum's consensus ³⁶ ³⁸ . It's especially useful for use cases like payments (immediate confirmation), rapid DeFi trading (multiple swaps within one block, updated live) and gaming/marketplaces that need snappy feedback ⁴⁰ .

Behind the scenes, achieving this requires tight integration: validators delegate their slot to a **Realtime Agent** (a special builder) via the ETHGas Whole Block market, that agent builds the block gradually and streams the state diff to the network while also submitting the final block at the deadline ¹⁹ ²⁰ . The diagram below illustrates this flow:

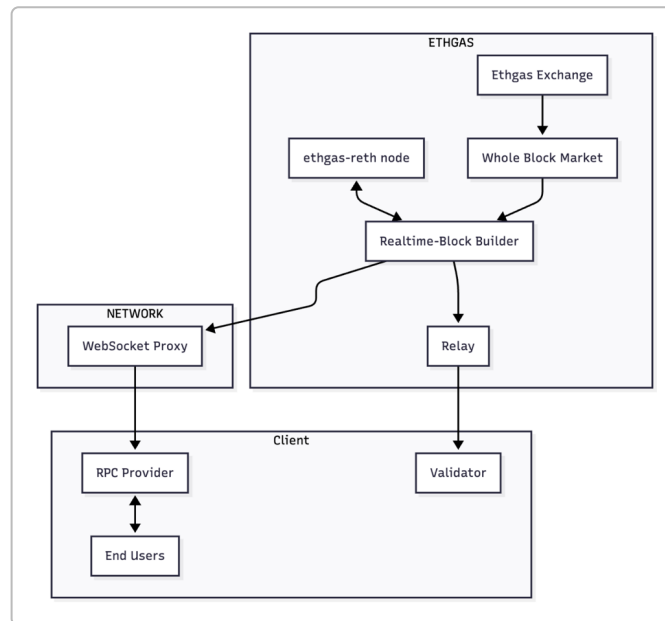


Diagram: How ETHGas integrates into Ethereum's block production. Validators can run "Commit Boost" to sell their upcoming block slots on the ETHGas Exchange. Buyers (traders, builders, or specialized realtime agents) bid in the Whole Block Market for a given slot ¹⁹ . The winning agent (if it's a realtime builder) constructs the block and streams preconfirmed state updates via a WebSocket proxy to RPC providers, reaching end-users with ~50-100ms latency ²⁰ . At the end of the slot, the finalized block is sent through the relay to the validator as usual ⁴¹ . This way, users experience near-instant transactions, and the block still settles on-chain normally. ³⁹ ³⁸

In summary, ETHGas's core mechanism is to **separate blockspace allocation from the immediate demand**. Instead of every transaction fighting in the mempool each block, a lot of the heavy hitters (like dApps or arbitrage bots) could plan their block usage ahead of time on ETHGas. This not only helps those participants (through cost certainty and speed), but potentially **stabilizes the public network** by smoothing out surges. The concept is somewhat analogous to **"priority boarding"** on an airplane – if enough people pre-book their spots, the remaining standby queue is more orderly. Of course, this introduces a new layer of market dynamics on top of Ethereum, which brings us to the question of **who wins, who loses, and what incentives are at play**.

Stakeholders and Incentives: Who Benefits (and Backs) ETHGas?

Winners / Beneficiaries: If ETHGas succeeds, several groups stand to gain:

- **Everyday Users (Retail):** Regular Ethereum users could enjoy **fee-less transactions at the point of use**. Instead of keeping ETH aside for gas or timing the network for low congestion, users would have apps or wallets that cover gas for them (possibly in exchange for a small subscription or simply as a perk). ETHGas's vision is that what today is a gas fee paid by a user becomes tomorrow's "*gas rebate*" – a reward or invisible cost handled by someone else ⁴² ⁴³ . This lowers the barrier for newcomers and makes Ethereum apps feel as seamless as Web2 apps (where you don't directly pay "bandwidth fees" for each click). Users also benefit indirectly from **more stable fees**; no sudden price spikes slowing the network for days.
- **dApp Developers and Protocols:** Applications get to **offer a smoother experience** (no scary gas prompts) which can attract and retain more users ⁴² . Moreover, with blockspace futures, *businesses can budget their gas costs*. For example, a DeFi platform running daily operations could hedge its gas expenses months in advance, ensuring that a spike in gas won't ruin their profitability or user experience ⁴⁴ . ETHGas essentially gives builders a tool to manage one of their biggest operational risks. Some protocols might even arbitrage gas costs – buying futures when cheap and using them during high-demand events. ETHGas reports that several Ethereum projects (e.g. liquid staking service ether.fi, restaking platform EigenLayer) have joined the **Open Gas Initiative** to sponsor gas for their users ¹³ . For them, covering user gas is a way to **boost adoption**. Rather than spending on ad campaigns, a protocol might spend funds to make usage free and thus more appealing.
- **High-Frequency Traders and MEV Searchers:** These are power users who generate lots of transactions (and often revenue) through arbitrage, liquidations, NFT sniping, etc. Such players highly value **speed and certainty**. For them, being able to *buy an entire block* or a guaranteed sequence in a block is like getting a private highway for their transactions. It eliminates the chance of getting outbid in the mempool and can protect against other bots (if you own the whole block, you won't be sandwiched by another MEV bot). ETHGas explicitly targets these users: whole blocks and "execution preconf" products are intended for **searchers, market makers, statistical arbitrageurs, and even centralized exchanges coordinating with DEXs** ²⁹ . In essence, it's a way for sophisticated traders to **pay a premium for guaranteed execution**, which could be worth it if the profit opportunity (MEV) in that block is high. These traders also benefit from gas hedging – e.g. an arbitrage bot can lock in gas prices during quiet periods, ensuring their strategy remains viable even if gas spikes later.
- **Rollup Operators and Large-Scale Users:** Ethereum Layer-2 networks and other big consumers of blockspace (like large organizations running on-chain operations) could use blockspace futures to stabilize costs. For instance, rollups like Arbitrum or Optimism must regularly post batches of data to L1 Ethereum. They could **pre-purchase block space** for their batch submissions during times of expected congestion (such as a popular token airdrop) to avoid paying extreme gas prices at those moments. In the long run, if blockspace becomes a *wholesale market*, L2s might contract with L1 validators for capacity the same way cloud services reserve server bandwidth. By doing so, rollups secure predictable costs and can offer steadier fees to their own users. (Notably, Ethereum's recent **proto-danksharding upgrade (EIP-4844)** created a separate *blob space* for L2 data, significantly reducing L2 posting fees ⁴⁵ ⁴⁶ . ETHGas-like markets could emerge for blob space as well, although ETHGas currently focuses on the main blockspace.)

- **Validators / Stakers:** Ethereum validators (and by extension, those who stake ETH) stand to **earn more** through ETHGas. Currently, validators earn priority fees from transactions and MEV via block builder auctions. ETHGas adds another revenue channel: selling future commitments. The team designed the marketplace to **“align highly with [validators’] interests”** ⁴⁷ and claims validators can **earn strictly more** than by just running MEV-Boost (Flashbots) alone ⁴⁸. By participating, validators get paid upfront for slots and can still collect normal fees, effectively monetizing blockspace *more fully*. To sweeten the deal, ETHGas plans a **Staking Incentive Program** where validators who engage with the system get a share of trading revenues ⁴⁹. This could boost staking yields, making Ethereum staking more attractive overall ⁵⁰. For honest validators, the additional risk is minimal – as long as they include the sold transactions, they aren’t slashed ⁵¹. Moreover, validators can choose how much collateral to put up and how many slots to “sell forward” (e.g. only commit some of their future blocks, keeping flexibility) ⁵² ⁵³.
- **Investors and the ETHGas Team:** ETHGas is venture-backed, so its **founders and investors** also expect to benefit if the platform gains traction. Polychain Capital led the \$12M funding, with other crypto VCs like BlueYard and Lafayette, and trading firms like SIG DT and Amber Group participating ¹⁰ ⁵⁴. They likely see upside in ETHGas through its native token and the fees it generates (more on the token shortly). The fact that **Ethereum Foundation researchers and core figures (like Vitalik and Justin Drake) have conceptually supported gas markets** ⁵⁵ ⁵⁶ may have given investors confidence that ETHGas’s approach will be well-received and potentially integrated into the Ethereum ecosystem long-term. ETHGas also announced it had **\$800M in initial blockspace purchase commitments** from “wholesale” participants (validators, builders, relays, etc.) ⁹ ⁵⁷, indicating significant early adoption – those committing likely expect to profit via the system (either by getting cheap blockspace or earning fees selling it).

Funding and Power Dynamics: Having heavy hitters on board (\$800M is a sizable chunk of blockspace reserved) raises questions about power concentration. If a few large players lock in a lot of future blockspace, could they **dominate Ethereum block access**? ETHGas asserts the market is open to all kinds of buyers (dApps, users, not just insiders) ⁸, but realistically, it will be mostly *professional market makers and protocols* using it, not casual users directly. There is a risk that **control over blockspace could centralize** among those who can afford to purchase it in bulk or who operate the infrastructure. However, one counterpoint is that even today, access to blockspace isn’t equal – MEV bots and wealthy users effectively “buy” their way to the front of the line with high fees. ETHGas formalizes this in a way that could actually democratize it (through a transparent market) or, if not designed carefully, favor insiders.

It’s worth noting ETHGas’s founder, **Kevin Lepsoe**, has a background in structured derivatives trading at Morgan Stanley ⁵⁸. The project is *literally financial engineering applied to Ethereum*. This brings sophisticated financial incentives but also potential **conflicts of interest**. For example, validators working with ETHGas might prioritize the ETHGas commitments over normal transactions, since those come with slashing risks and pre-arranged fees. That could marginalize the “free” public mempool if most block space gets pre-sold (imagine regular transactions always being second-class, only using leftover block space). On the other hand, ETHGas commitments are short-term (at most ~64 blocks ahead for full blocks ⁵⁹), and any unused block capacity still gets filled with normal txns, so the public transaction pool should still find room, just with less volatility.

Who might lose or face disadvantages? If ETHGas markets take off, **opportunistic arbitrageurs who rely on gas price swings** could lose some edge (because futures would smooth out extreme fees). Also, users who *don’t* participate in the new system might feel left out – e.g. an average user not using a gas-sponsored wallet could still face variable fees, and possibly higher ones if blockspace gets largely taken up by those with pre-confirmations (though ETHGas’s goal is to reduce overall volatility, not necessarily

average cost). **Miners/validators who don't join** might earn less if much MEV and order flow moves to ETHGas-coordinated blocks (but since Ethereum is now Proof-of-Stake, most validators will likely opt in if it's profitable, or risk underperforming).

Finally, the **Ethereum protocol itself** must balance this innovation with fairness. There could be social questions: should a lot of blockspace be locked by private contracts off-chain? Vitalik and others seem to support *trustless* on-chain markets ⁴, meaning ideally these mechanisms would be baked into Ethereum or at least operate transparently. ETHGas is working “in close alignment” with the Ethereum Foundation ⁵⁵, suggesting they intend to keep it aligned with Ethereum’s decentralization values, but it’s a fine line between making Ethereum more like a predictable utility versus introducing Wall-Street-like complexity.

Risk Surfaces and Red-Team Analysis: What Could Go Wrong?

Every new system comes with risks, and a blockspace futures market is no exception. Here are key **attack surfaces and challenges** that ETHGas and its users will need to watch:

- **Validator Manipulation of Gas Dynamics:** One concern raised is that introducing futures could incentivize validators to *manipulate the base fee mechanism*. For example, if a validator sells a future guaranteeing low gas prices, they might try to intentionally under-fill blocks to push the base fee down ⁶⁰. Conversely, someone shorting (betting against) gas price might try to spike it. Kevin Lepsoe himself cautioned that on Layer-1, validators might **game Ethereum’s base fee** if not mitigated ⁶⁰. Ethereum’s base fee (under EIP-1559) adjusts 12.5% up or down depending on block fullness. A colluding builder could, in theory, leave blocks half-empty to drop fees (or overstuff with junk to raise fees) if they had a large futures position. This risk is partly why **ETHGas suggests focusing on L2s or a separate layer** for the futures market, where such manipulation would be less harmful ⁶⁰. The design with collateral and slashing also discourages validators from not filling blocks – they’d lose money if they breach commitments. Still, careful monitoring is needed to ensure validators don’t exploit subtle tactics to profit on both the futures and the on-chain fees.
- **Market Manipulation & Speculation Risks:** Any derivatives market can be manipulated, especially if illiquid. Blockspace futures could be vulnerable to **whales cornering the market** – e.g. a large trader could buy up a lot of block capacity for a critical timeframe (say, around a big token launch or an NFT drop) and either **resell it at exorbitant prices** to frantic users or simply exclude others by owning all the space. This could become a new form of **MEV**: instead of racing in a one-block auction, a savvy actor could secure whole blocks in advance when demand is anticipated. This is partially by design (market makers will indeed try to anticipate demand), but it could disadvantage those who are unaware of the futures market or can’t afford to participate. There’s also the scenario of **false demand signals** – someone could drive up futures prices (making it look like blockspace will be scarce), which might panic others or affect decisions in the ecosystem. However, since blockspace is a perishable good (you can’t “store” an unused block), attempts to hoard it for profit have limits; if you buy a block and don’t use it, you lose that opportunity cost. ETHGas does allow buyers to leave block space unfilled if they choose, but by default any unused gas can be filled by the block builder with other transactions ⁶¹, so it won’t go entirely to waste. Still, **short-term squeezes** or speculative swings in futures prices are possible, especially early on.
- **Sybil and Identity Attacks:** One expert pointed out that if a significant premium is placed on future block reservations, it might encourage **Sybil attacks in the identity layer** ⁶². For

instance, if only “reputable” players are allowed certain privileges or if some kind of identity staking is needed (this isn’t currently part of ETHGas, but imagine if in the future only verified dApps can buy cheap blockspace), then malicious actors might create fake identities to exploit any allocation mechanisms. More straightforward, a manipulator could pose as many buyers to either jack up prices or overload the system. ETHGas will likely require accounts to post collateral and have KYC/agreements (since big validators and traders are involved), which provides some friction against Sybils.

- **Platform Security and Smart Contract Bugs:** ETHGas relies on off-chain software (commit modules, exchange servers) and on-chain contracts (for collateral management, slashing, etc.). These introduce **new attack surfaces**. A bug in the **collateral contract** could be catastrophic – e.g. if slashing doesn’t occur when it should, a validator could sell promises and fail to deliver with no penalty. Or if the exchange matching engine has a flaw, an attacker could exploit it to steal collateral or disrupt the market. ETHGas posting its own bond for liveness ²⁴ is reassuring, but doesn’t eliminate the risk of technical failure. The complexity (just look at the multi-step process in the docs for how orders and bundles are submitted ⁶³ ⁶⁴) means there are many points where something could go wrong. From a “red team” perspective, one would scrutinize the **slashing conditions** – e.g. what if a validator includes the transactions but reorders them unfairly? Or includes them but in a way that violates some expectation? The system has to define “default” conditions precisely ²³ . Per the docs, if *any* sold pre-confirmation isn’t included exactly as promised, it’s a slashable event ⁶⁵ ⁶⁶ . But ensuring detection of that on-chain is non-trivial (they plan to check the block’s content after finalization and then credit affected buyers from slashed funds) ⁶⁷ ⁶⁸ . Attackers might find edge cases – for example, could a validator include a transaction but in a way that renders it ineffective (perhaps by manipulating state elsewhere)? These are highly technical questions, but they illustrate that **economic guarantees are only as good as the code** enforcing them.

- **Increased Complexity (for Users and Devs):** While ETHGas’s goal is to *simplify* life for end-users (no gas headaches), achieving that requires a **lot of complexity behind the scenes**. Developers who want to integrate gas sponsorship or use blockspace futures have to interact with new APIs, manage futures positions or preconfirmations, and possibly deal with **margin/collateral** if they themselves start speculating. There’s a risk that not all projects will have the expertise to do this correctly, leading to mistakes. For instance, if a dApp misjudges and doesn’t buy enough blockspace for an event, its users might still face congestion. Or if it buys too much and overpays, that’s wasted funds (which could harm the project’s finances). Users might also become **confused by varying experiences** – one app might be gasless because it uses ETHGas, while another still requires manual gas, potentially leading to misconceptions (“I never pay gas on X, why is Y charging me?”). Moreover, in a gasless model, users **lose the signal of network demand**; today high gas alerts a user that the network is busy (and maybe they should wait or pay more for priority). If everything seems free, users could submit lots of transactions without realizing there’s a crunch behind the scenes – possibly leading to sponsored transactions queuing up or failing if sponsorship budgets are exceeded. In short, **abstracting gas adds opacity** that could backfire if not communicated. It will be important for wallets and apps to handle errors gracefully (e.g. if the gas sponsor runs out of credits, does the user then suddenly get a gas prompt or a failure?).

- **Financial Risk & Cascading Failures:** Introducing futures means **participants take on new financial risks**. For example, if a trading firm commits to buying 100 blocks of capacity and then a black swan event occurs (say Ethereum usage drops drastically or another technology makes blockspace less needed), they could incur losses on those futures. Conversely, validators are putting up collateral – if a bug or network issue causes them to miss slots, they get slashed

financially. A concern is **liquidity and defaults**: ETHGas uses 100% initial margin for futures ³⁰, meaning no leverage, which is conservative. But if someone's collateral is insufficient (say gas prices move wildly and they can't top up for a short bet), how smoothly will the system handle liquidations or defaults? Could a large default cause shock waves (for example, a validator defaulting on many commitments might lead to lost transactions, user frustration, and questions of compensation)? The design tries to localize this (slashed collateral pays the affected buyers ⁶⁹ ⁷⁰), but extreme events might exceed collateral – at which point some promises would be broken. That's a new kind of failure mode for Ethereum: today, a transaction might fail or get stuck, but with futures, you could have *entire batches* of promised transactions fail if, say, a major builder node went down.

- **Effects on Fee Market and User Fees:** Another potential risk is that **futures could distort the normal fee market signals**. If a large portion of blockspace is consistently sold in advance at a fixed price, the remaining on-demand space might see either lower volatility (good) or weird oscillations. For instance, the presence of many privately pre-filled blocks could mean that occasionally, the public mempool blocks are under-filled (because some validators are busy only including their pre-sold transactions). This scenario was observed with private orderflow: “vanilla” public blocks sometimes come out only 15% full, dropping the base fee, then the next block (with withheld private txs) comes 100% full, spiking it ⁷¹ ⁷². If not balanced, such **alternating fullness** could cause gas price whiplash block-to-block. ETHGas's coordination might actually reduce this by smoothing inclusion over slots, but it's something to monitor – *volatility could migrate to shorter timeframes*. The team will likely refine parameters (how far ahead you can buy, how much of a block can be pre-sold, etc.) to avoid pathological cases.

In summary, the **attack/risks surface is broad**: from economic manipulation to technical bugs to user perception. The “first defense” against these is transparency and gradual testing. Encouragingly, ETHGas is rolling out with major ecosystem players and presumably auditing its contracts. Ethereum's community is also aware of these trade-offs – Vitalik's call for gas futures came with no guarantee it's easy or risk-free, and indeed the idea has “**split experts**” ⁷³. Some believe it's necessary for the next phase of Ethereum, others worry it adds undue complexity for not enough benefit ⁷⁴. As ETHGas enters the wild, it will need a strong **red-team mindset**: assume people will try to break or game the system, and be ready with safeguards (like the slashing mechanism, circuit breakers on the exchange, monitoring for abuse, etc.).

Narrative vs. Reality: Is a “Gasless” Ethereum Truly Free?

ETHGas markets itself as enabling a “*gasless*” future, conjuring an image of using Ethereum without paying any fees. The reality is a bit more nuanced. **Ethereum will always require gas fees at the protocol level** – it's a fundamental security feature to prevent spam and allocate limited blockspace. ETHGas doesn't eliminate those fees; it **redistributes and abstracts them**. Instead of individuals each paying gas for their own transactions, larger entities (protocols, services, or the ETHGas system itself) handle the gas in aggregate, ideally in a more efficient way.

Here's what stays the same and what changes in a “gasless” paradigm:

- **Fees still exist (and ETH is still spent/burned):** Even if a user doesn't personally send 0.005 ETH for a transaction, that ETH must come from somewhere. If a dApp sponsors the gas, the dApp is paying the network fees on the user's behalf (likely using ETH it acquired or reserved earlier). Ethereum's base fee will still be burned in each block ³², maintaining the economic properties of EIP-1559. ETHGas doesn't avoid the base fee burn – it just means whoever bought the

blockspace would factor that cost in. For example, if the base fee is 20 gwei and you pre-bought a full block, you effectively are covering that base fee for all included transactions (plus perhaps a tip to the validator). Those costs might be bundled into what you paid upfront. So from Ethereum's perspective, fees are being paid and ETH is burned as usual; from the user's perspective, **they didn't have to pull out their wallet to pay it at transaction time.**

- **Who pays and where do the fees go?** In ETHGas's model, **wholesale buyers pay fees to validators**, and validators (proposers) still ultimately get compensated for including transactions (through the blockspace sale). So validators continue to earn rewards (and even higher via ETHGas's fees) ⁴⁹. The difference is the payment is negotiated in advance rather than via tips in the mempool. If a dApp is the buyer, then effectively **the dApp's treasury pays the gas**. That money might come from the dApp's revenue, token reserves, or perhaps venture funding allocated to user growth. ETHGas itself likely **takes a cut** as a marketplace fee (similar to how an exchange charges trading fees). In fact, ETHGas mentions a portion of *trading revenue* will go to stakers in their ecosystem ⁴⁹, implying that the platform charges fees on these blockspace trades and those fees can be shared or used as incentives. Thus, ETHGas will profit by **facilitating gas fee transactions** – essentially becoming a middleman for blockspace. Over time, one could imagine ETHGas (or its decentralized successor) taking a tiny percentage of each block bought/sold, analogous to an exchange fee.

- **ETHGas's Token and Profit Model:** ETHGas introduced a native token called **\$GWEI** in January 2026 ⁷⁵. \$GWEI is primarily a **governance token** to let the community vote on protocol changes and treasury use ⁷⁶. However, such tokens often have indirect value accrual – for example, fees from the protocol could flow into a treasury that \$GWEI holders control or receive via buybacks, etc. The token's supply is 10 billion, with significant allocations likely for the team, investors, and ecosystem incentives (e.g. an airdrop to early users) ⁷⁷. Indeed, ETHGas announced a **"Gasless Future" community airdrop** using a **"Proof of Pain"** concept – meaning the more gas you've spent historically (and the more you're verified as a real user), the more \$GWEI tokens you might receive ⁷⁸. This cleverly targets real users who suffered high fees in the past, effectively **rewarding them with a stake in the new system**. From a profit standpoint, ETHGas's team and backers presumably hold a share of \$GWEI, so if the platform is widely adopted, the token could appreciate in value, benefiting them. Additionally, if ETHGas runs any infrastructure (like its own block builder or relay), it might earn a portion of MEV or fees directly.

- **Is "gasless" misleading?** It's somewhat a **marketing shorthand**. A more accurate phrase might be *"user-facing gasless"* or *"gas-abstracted"*. There have been "gasless" solutions before (for example, some wallets use meta-transactions where a relayer pays the gas and the user pays the relayer in a different way), but ultimately someone pays. ETHGas's narrative is optimistic: they say **every transaction can be instant and predictable with invisible gas fees** ¹⁷. This paints a future where using Ethereum feels like using the internet – you're not dropping a coin in a meter for every click. However, consider that even in the internet analogy, you *do* pay for bandwidth via your ISP, it's just a flat subscription rather than per webpage. Similarly, Ethereum users might in the future pay a subscription to a service that covers all their gas, or perhaps watch ads or provide some value in exchange. ETHGas is just enabling dApps to build those models by removing the volatility and uncertainty of gas costs. **So "gasless" doesn't mean Ethereum is free; it means the costs are handled in bulk and possibly passed on indirectly (through token economics, subscriptions, or higher layer-2 fees).**

- **What fees remain and new fees?** If ETHGas is widely used, **base fees and priority fees** on L1 still remain as they are – but much of those might be transacted via the ETHGas marketplace rather than the public mempool. On layer-2s, users might see *zero* fees if the L2 operator covers

them (some rollups have done promotional gasless periods). But that cost goes to the L2 operator's balance sheet (which in turn comes from investors or token emissions). So one could argue the fee is paid via **inflation of a token or via opportunity cost**. One *new* type of "fee" introduced by blockspace markets is possibly **financing cost**: if you prepay for blockspace, that's capital tied up until you use it. For instance, an app might spend \$1M to secure blockspace for the next 6 months. The *time value* of that money is a cost (they could have invested it elsewhere). If markets are efficient, futures prices will reflect expected gas plus a small premium for convenience/stability (and perhaps interest). So, users may indirectly pay a bit more for the convenience of stability – like an insurance premium. ETHGas claims transactions become **"budgetable"** ¹¹ ⁷⁹, which is a huge benefit for planning, but that insurance might not be free. It will be competitive though: if premiums are too high, many might just stick to pay-as-you-go gas.

- **How does ETHGas itself make money?** As mentioned, likely through **fees and token value**. If ETHGas is run as a protocol, it could charge a tiny fraction of each trade. For example, if someone buys a block for 0.1 ETH, perhaps 0.001 ETH goes to a fee pool. With high volume (imagine many blocks per day traded), this could accumulate. Their planned **staking program** suggests those who stake \$GWEI might get a share of trading fees, which implies a portion of fees is set aside as rewards ⁴⁹. This is similar to how some decentralized exchanges distribute fees to token stakers. In short, ETHGas's profit model likely aligns with creating a **liquid marketplace** – the more blockspace traded, the more fees earned. It's also worth noting that by making Ethereum more usable (gasless, instant), **the overall pie can grow** – more transactions and activity could occur, indirectly increasing demand for blockspace and ETH (which could, in turn, raise gas prices long-term, but hopefully in a more controlled way). ETHGas's investors are probably betting that *if* blockspace becomes a major commodity, they are effectively investing in the "Chicago Mercantile Exchange of Ethereum" and will own a piece of that exchange.

Narrative vs reality check: Phrases like **"Realtime, Gasless Ethereum"** are inspiring, but we should parse them realistically. *Realtime* is partially achieved through clever engineering (pre-confirmations), but finality still depends on Ethereum's base protocol (currently ~12s, maybe ~6s in the future with upgrades). *Gasless* is achieved through **gas abstraction**, which has trade-offs (someone eats the cost, and that someone will expect something in return, whether loyalty, data, or fees elsewhere). Importantly, **gas fees are not reduced to zero by ETHGas** – they may be reduced *somewhat* by smoothing out inefficiencies (no more overbidding in spikes, etc.), but users might pay for them in other ways (like holding a token or paying a fixed monthly fee to a provider). Also, there will always be some form of **queue or rationing** – if demand wildly exceeds supply, even blockspace futures will get expensive (market forces). "Gasless" doesn't mean unlimited throughput; it means the *cost to the user* is hidden.

One should also consider **who carries the burden** in gasless models. If every dApp is sponsoring gas, their sustainability comes into question. They would need revenue streams (perhaps taking a small cut of user transactions or having a token economy) to afford those gas costs. Some might fail if they miscalculate. In DeFi, for example, **sponsored gas might encourage more transactions (good)** but also could lead to spam or overuse if not metered (bad if the sponsor foots a huge bill). It's a delicate balance: *free* services often have limits or monetization elsewhere.

In reality, we might see a **hybrid fee model** emerge. Perhaps basic transactions are covered (like a few per day), but heavy usage requires either holding tokens or still paying something. ETHGas gives the tools to experiment with these models.

Conclusion: Toward Ethereum's Next Era – Opportunities and Vigilance

ETHGas represents a bold evolution in how we think about using Ethereum. By converting **blockspace into a tradeable, reservable commodity** ⁸, it strives to bring economic efficiency and predictability to a resource that has historically been chaotic. If successful, it could **usher in an era of “invisible” gas fees**, where using Ethereum feels as smooth as using any web app – no more fumbling with Gwei settings or budgeting for unpredictable surges. Transactions could be confirmed in milliseconds via off-chain pre-confirmation, fulfilling the dream of a “*realtime Ethereum*” ³⁶ that doesn't sacrifice security.

However, as we've detailed, this comes with **significant complexity and new risks**. Ethereum's strength has always been in credible neutrality and simplicity at the base layer. Introducing futures markets and sidecar block-building must be done in a way that **preserves fairness and decentralization**. It will be crucial to monitor how ETHGas's early adoption plays out. Some questions to watch in 2026 and beyond:

- Will enough validators participate to make the futures market robust, and will their behavior remain honest under these new incentives?
- How will gas prices on the **spot market** (regular transactions) react? Will we truly see smoother prices and better UX, or will unforeseen consequences emerge (like new forms of MEV concentration)?
- Can the **governance token \$GWEI** successfully decentralize control of this critical infrastructure, or will a few entities dominate the blockspace market?
- Will users actually feel the benefits (e.g. mainstream apps adopting gas sponsorship) or will it remain an obscure tool for power users?

The narrative is certainly forward-looking – phrases like “*split the atom of Ethereum blockspace*” are used to signify a big breakthrough ⁸⁰. In reality, it will likely be iterative: we may see **gradual rollout** – perhaps specific use-cases (like certain DeFi platforms or NFT marketplaces) using ETHGas to guarantee transactions during launches or promotions, demonstrating the value. As comfort grows, more protocols might join the Open Gas Initiative, making “user-pays-gas” an exception rather than the norm.

One can draw a parallel to how **cloud computing** evolved: initially, everyone ran their own servers (like everyone managing their own gas fees). Over time, companies began reserving capacity, using spot instances, etc., to optimize costs – analogous to blockspace futures. Eventually, many users don't think about the server at all (like a user not thinking about gas) because that is abstracted by a provider. Ethereum could be heading down a similar path, with ETHGas and similar projects being the brokers of capacity.

In conclusion, ETHGas is a fascinating and ambitious attempt to solve one of Ethereum's longest-running pain points. It has strong backing and timing on its side – coming at a moment when Ethereum's core protocol is also evolving (sharding, danksharding, etc.) to scale capacity. A “**gasless Ethereum future**” is not here yet, but we now have a clearer picture of how it might be constructed. As the technology and market mechanisms mature, the general public might one day use Ethereum apps without ever worrying about gas, even if under the hood a complex market of blockspace trading hums along. Reaching that point will require continued **transparency, careful risk management, and community trust**. This first-defense analysis highlights both the promise and the pitfalls; moving forward, researchers and developers will need to remain vigilant (red-teaming the design, refining the economics) to ensure that the reality of gasless, real-time Ethereum lives up to the optimistic narrative in an equitable way.

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